Ethanol Blending in Gasoline – Bulgaria

September 2025

U.S. GRAINS & BIOPRODUCTS COUNCIL

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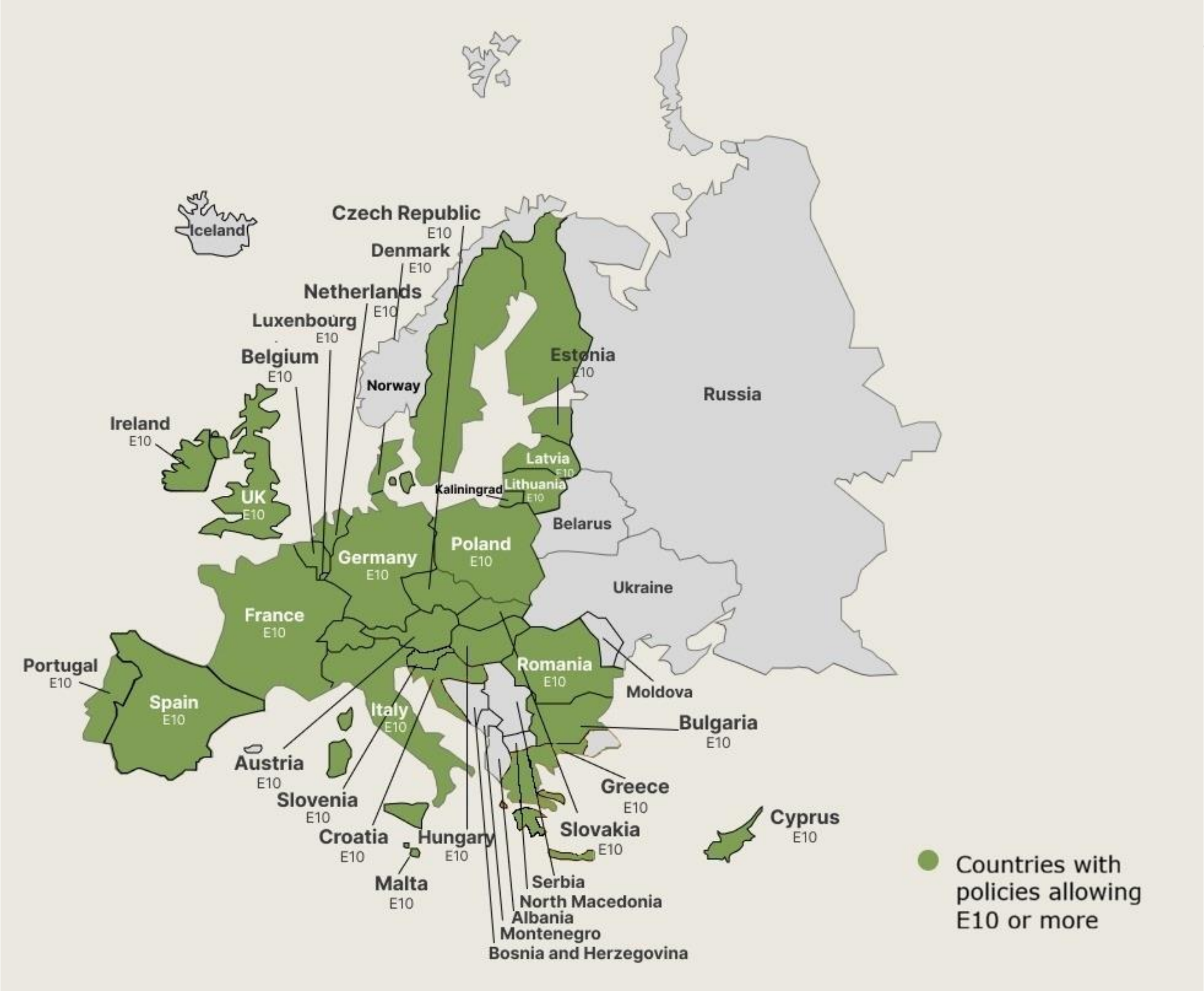
Ethanol Blending in Europe

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

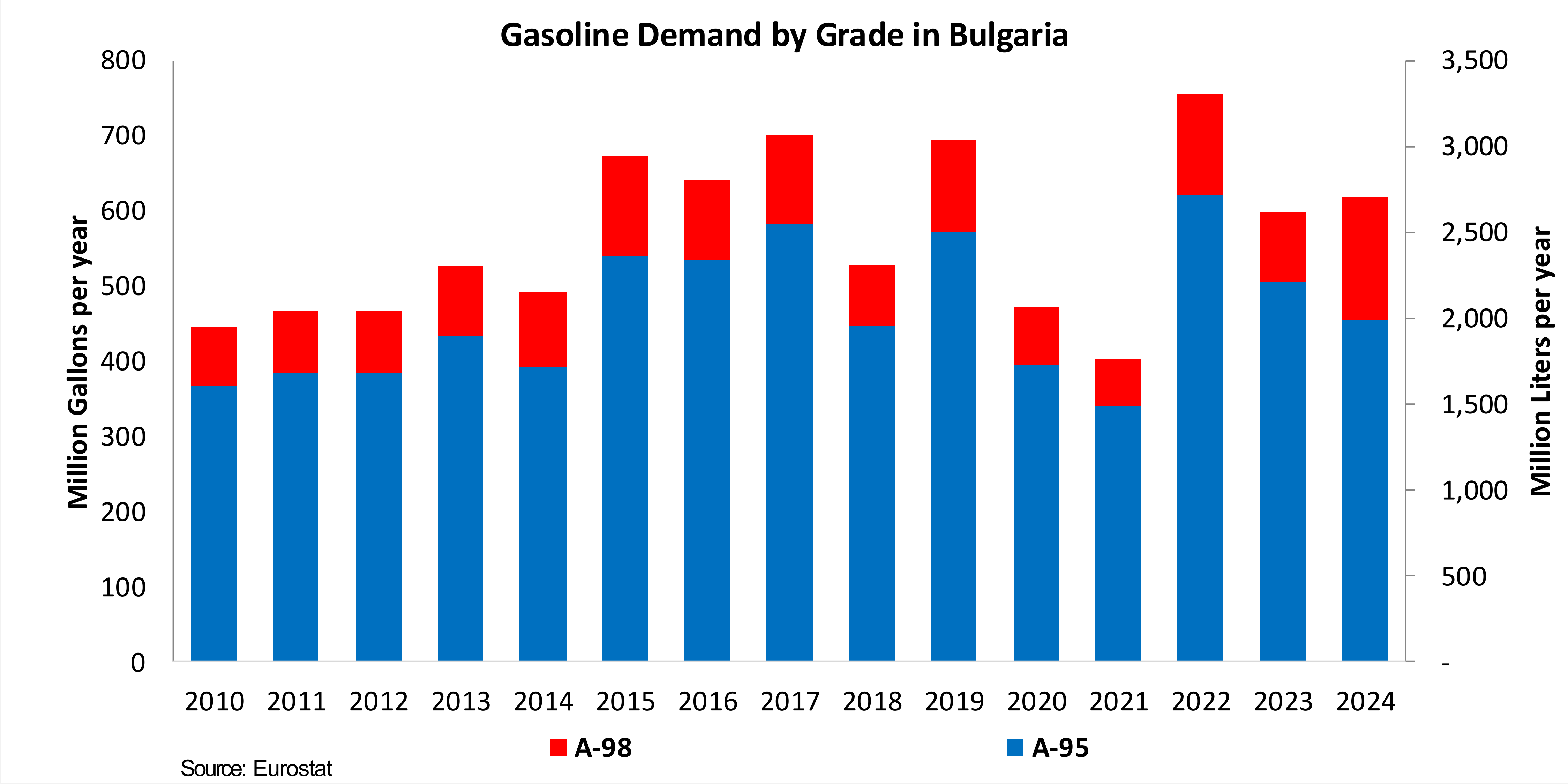
Ethanol Gasoline Blending in Bulgaria

In 2024, gasoline consumption in Bulgaria was 2,342 million liters (619 million gallons) and growth rate was -2.3%. Gasoline production and net imports were 2,199 and 143 million liters (581 and 38 million gallons) correspondingly. Bulgaria complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.5 RON.

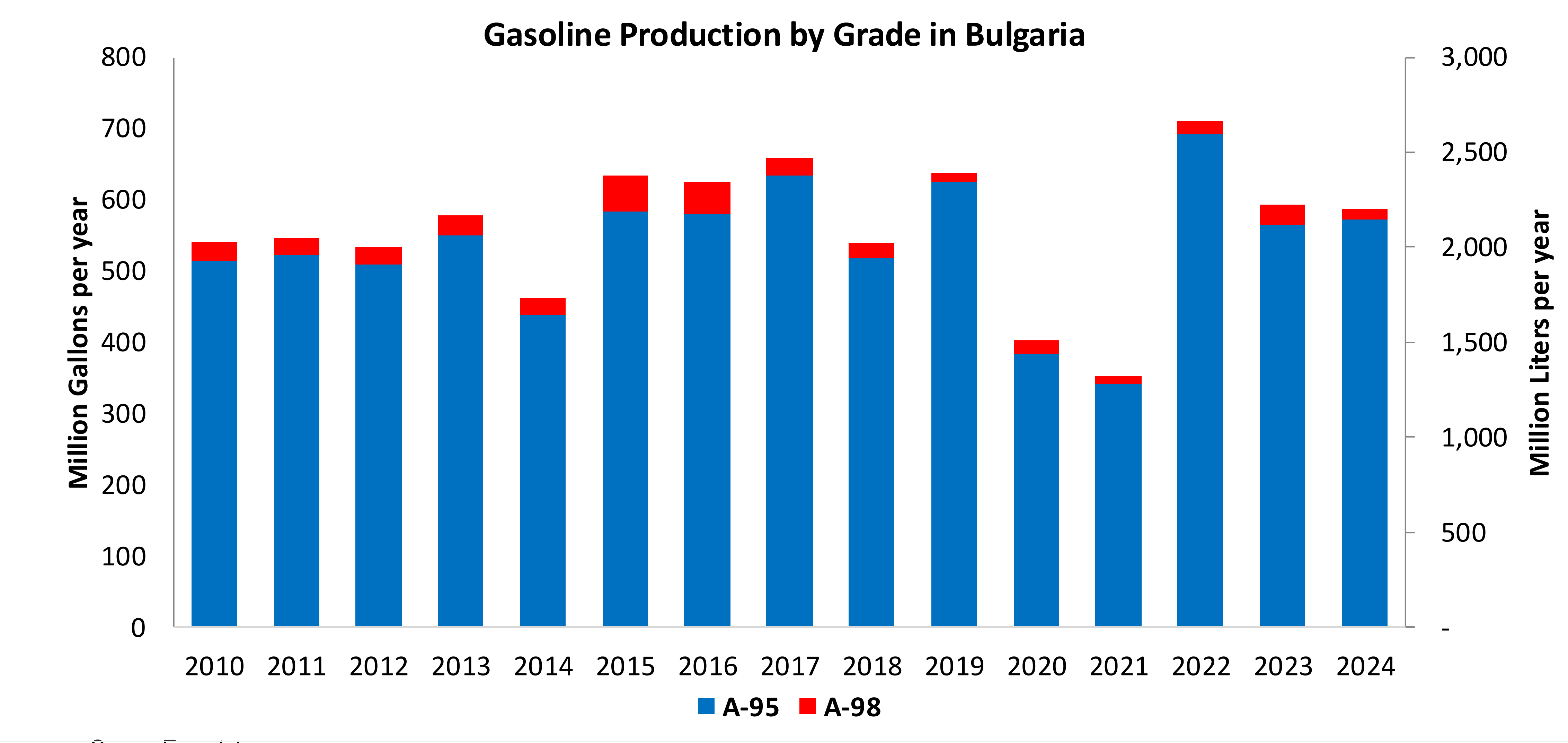
In 2024, ethanol consumption in Bulgaria was 54 million liters (14 million gallons) representing 2.3% of gasoline demand. Ethanol production and Net Imports were 40 and 14 million liters (14 and 4 million gallons) correspondingly. Ethanol sources are Corn 45% and Cereals 55%.

Source: Eurostat, European Commision

Gasoline Demand in Bulgaria

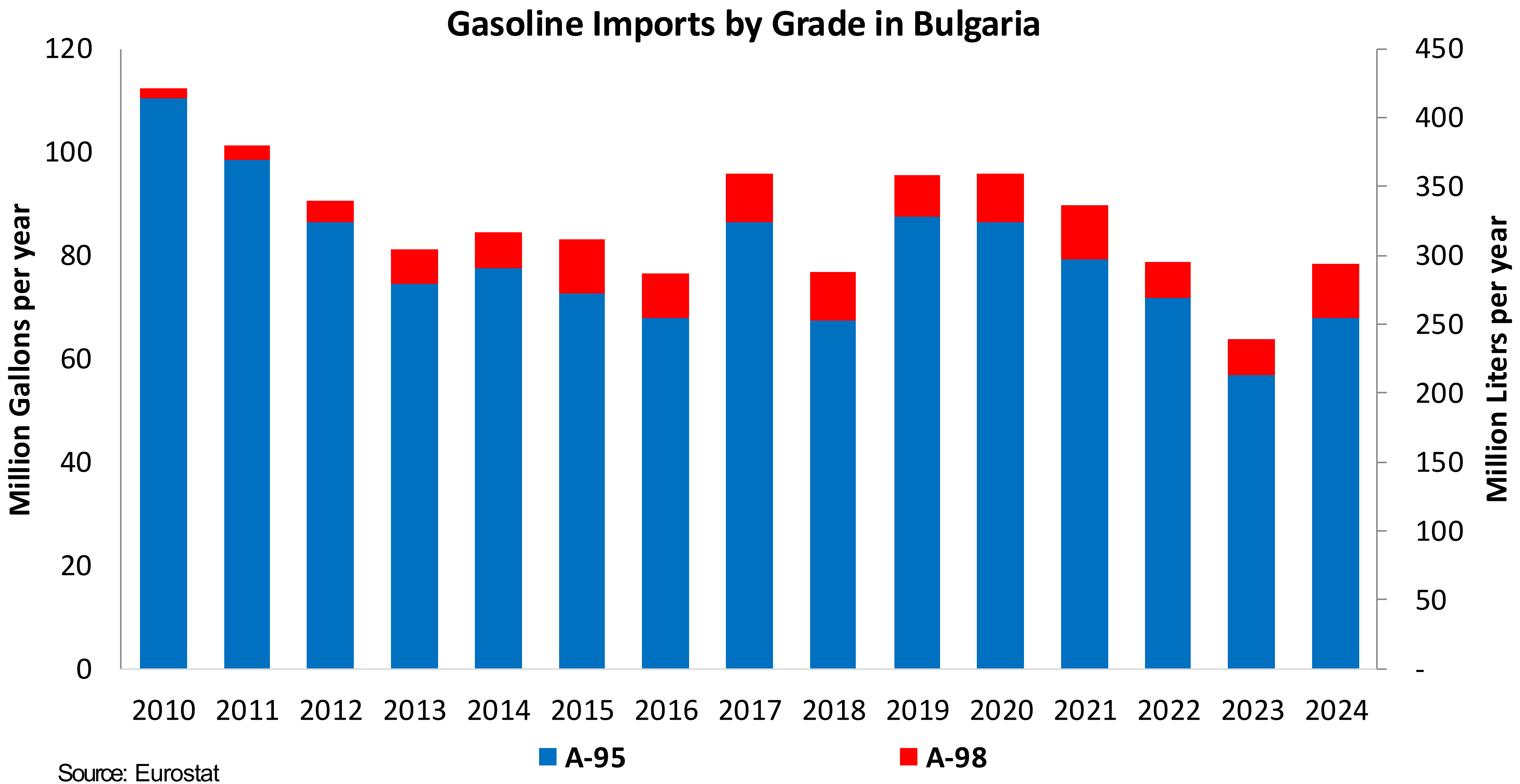


Gasoline Production in Bulgaria



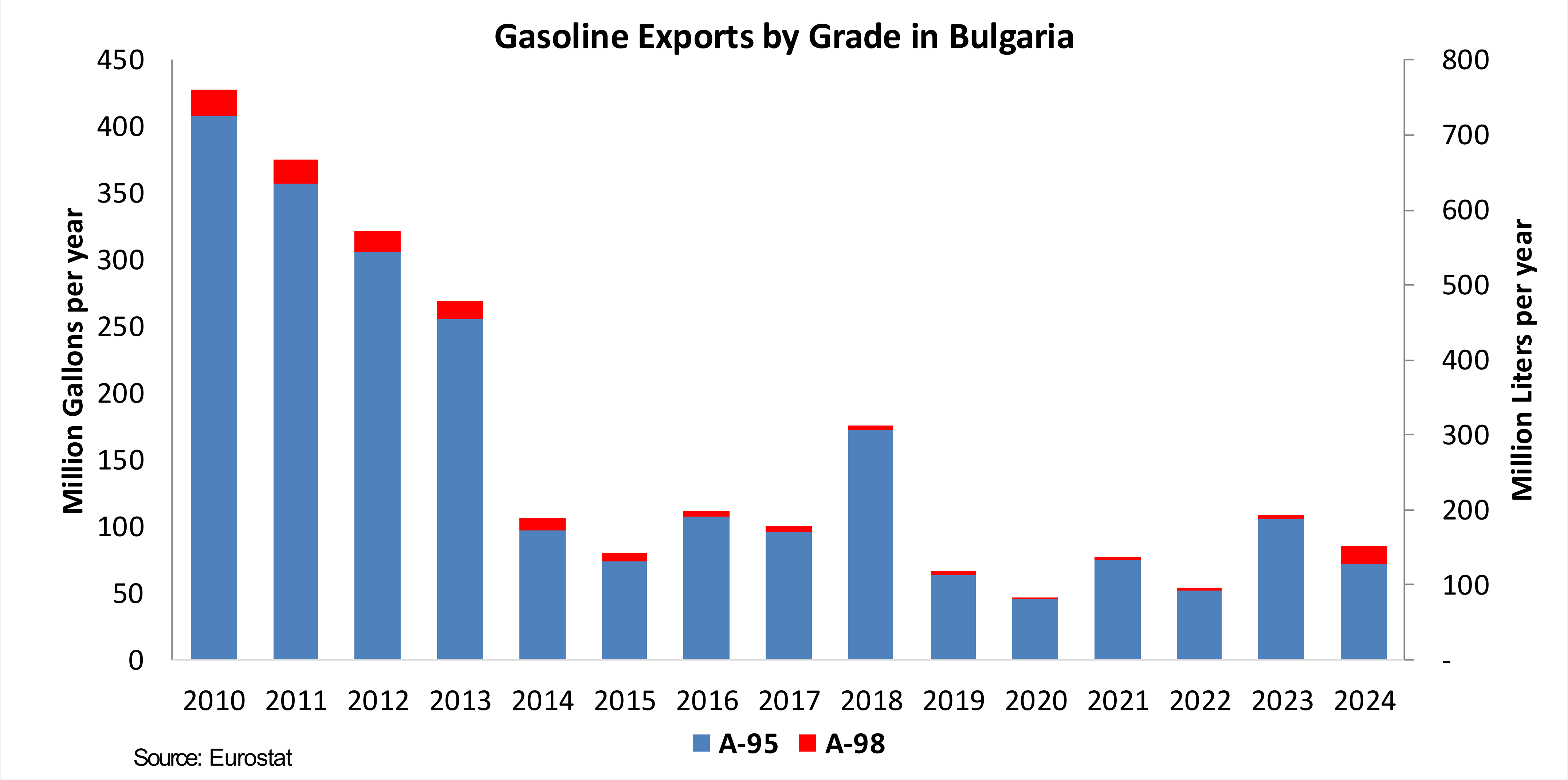
Source: Eurostat

Gasoline Imports in Bulgaria

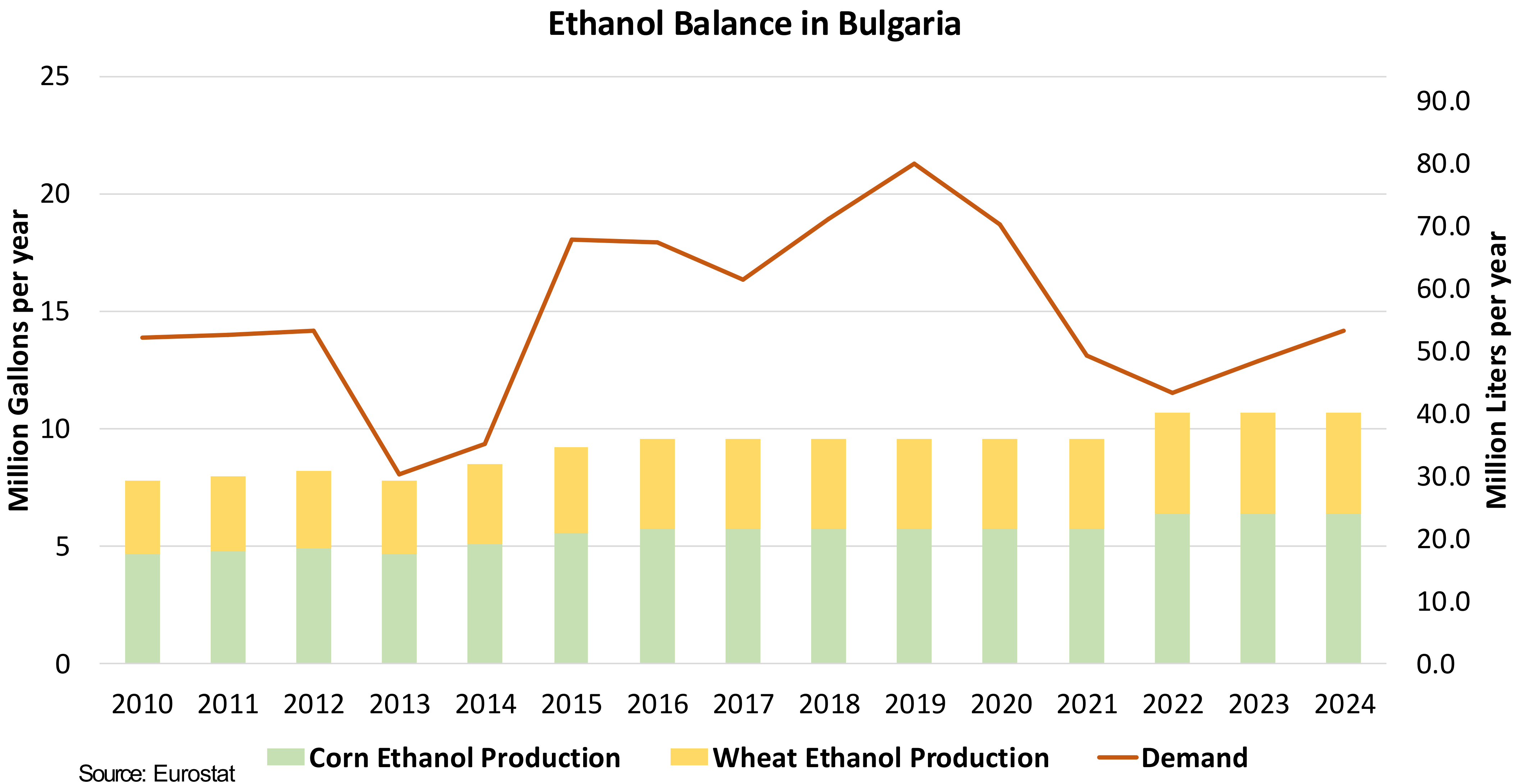


Source: Eurostat

Gasoline Exports in Bulgaria



Ethanol Balance in Bulgaria



Source: Eurostat

Gasoline Quality in Bulgaria

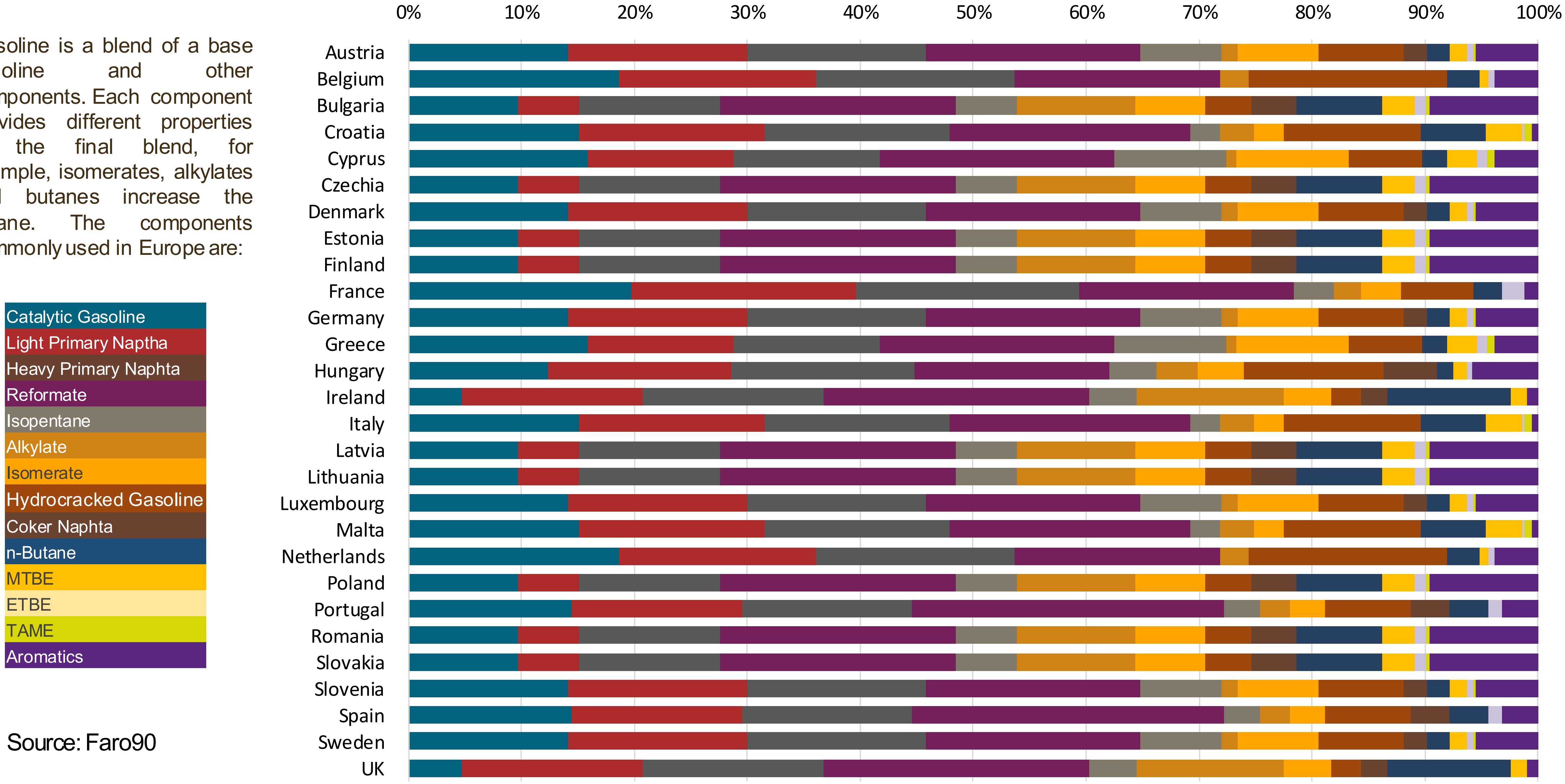
Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	1.0							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Gasoline Component Blending in Europe

Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:



Source: Faro90

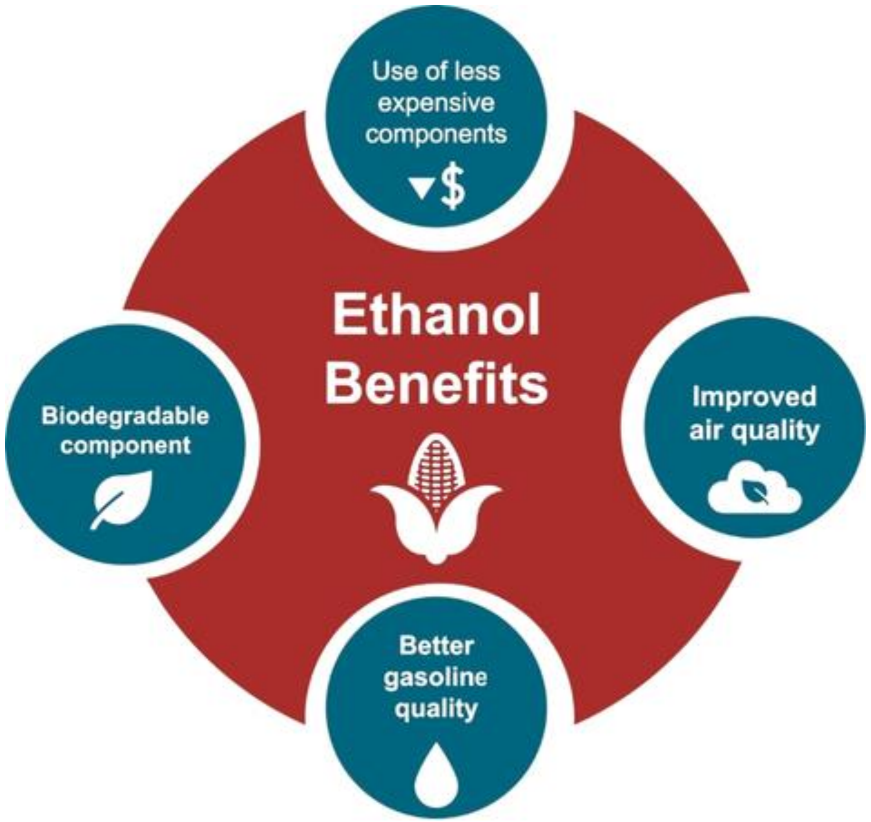
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component.The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives.To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

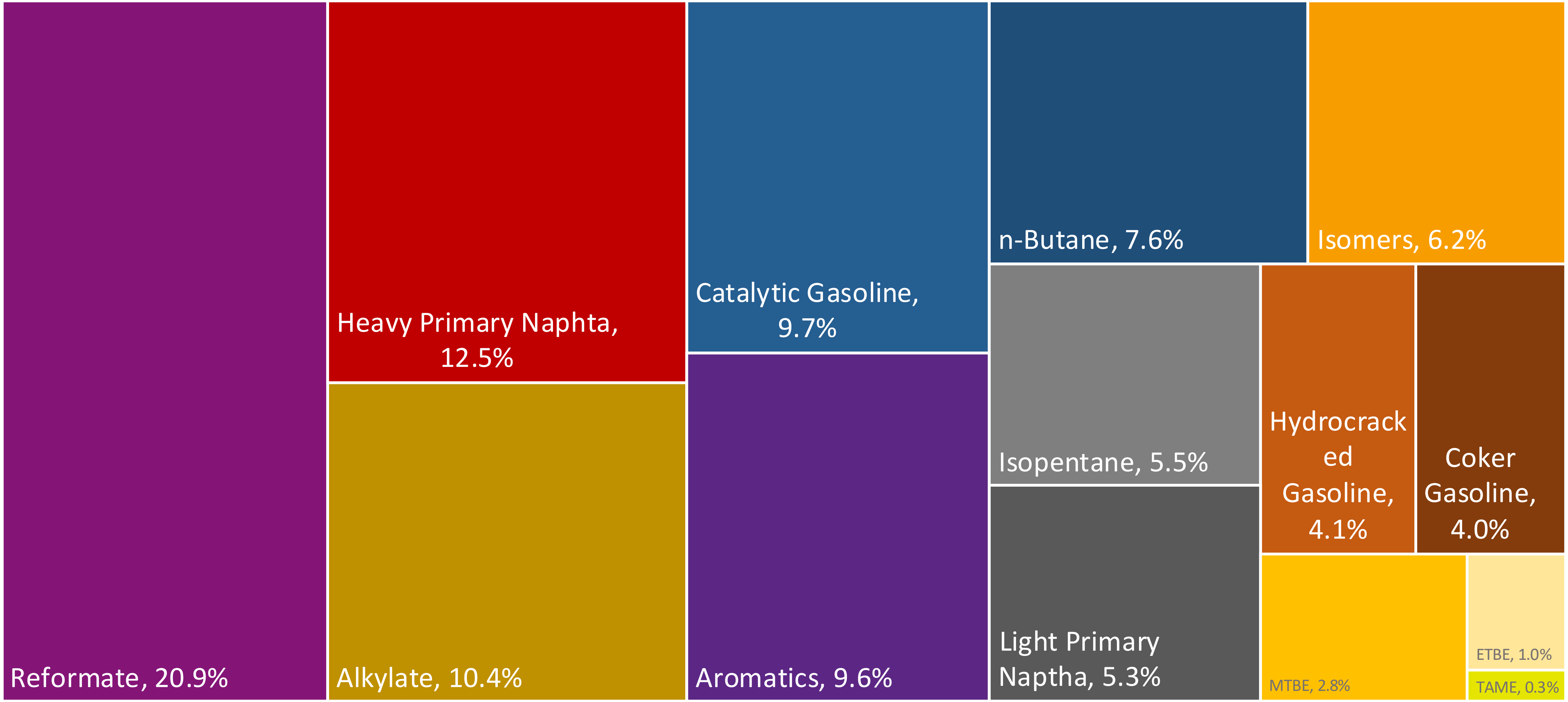
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

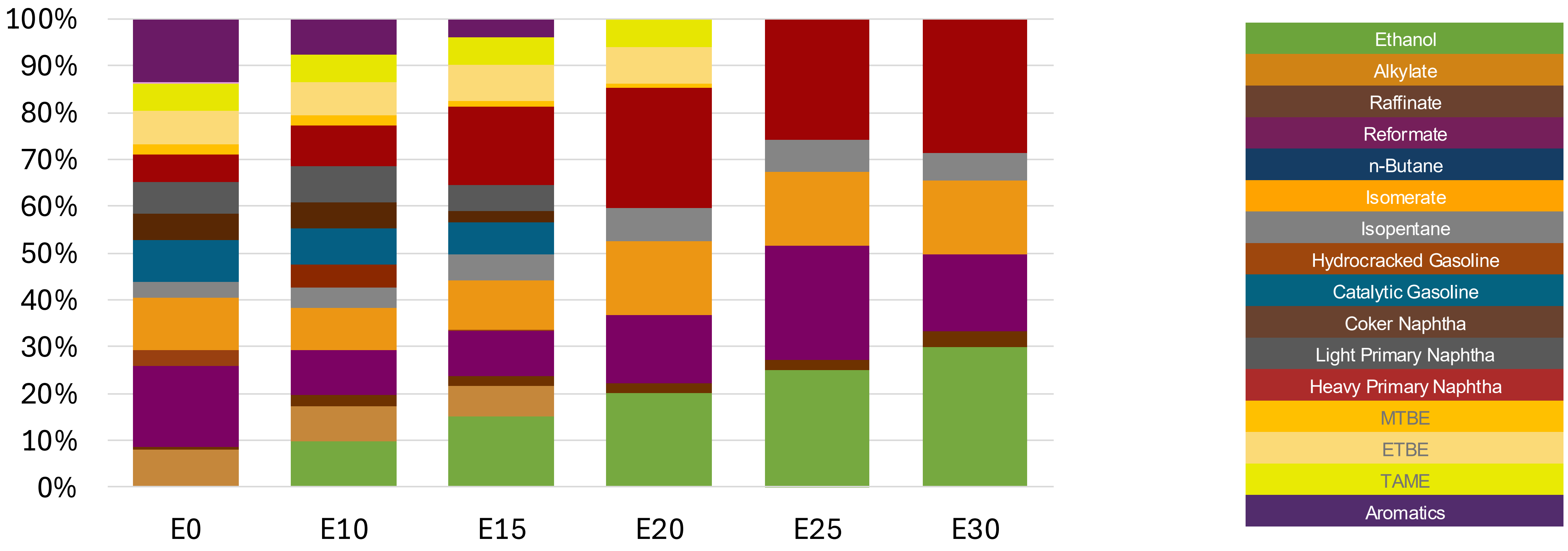


Bulgaria Available Blending Components



Source: Faro90

Bulgaria – Gasoline 95 – Constant Octane

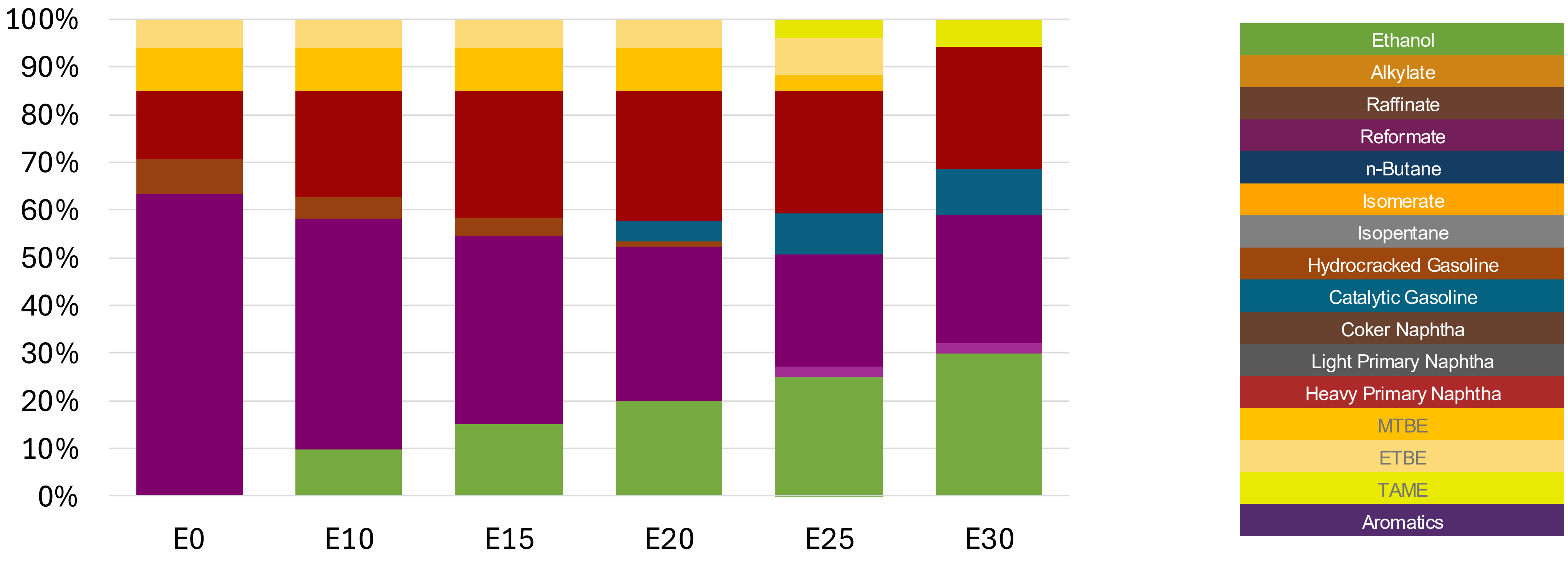


Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US$/gal)	$ 2.294	$ 2.156	$ 2.056	$ 1.941	$ 1.895	$ 1.842

Average CIF 2024 Prices

Source: Faro90

Bulgaria - Gasoline 98 - Constant Octane

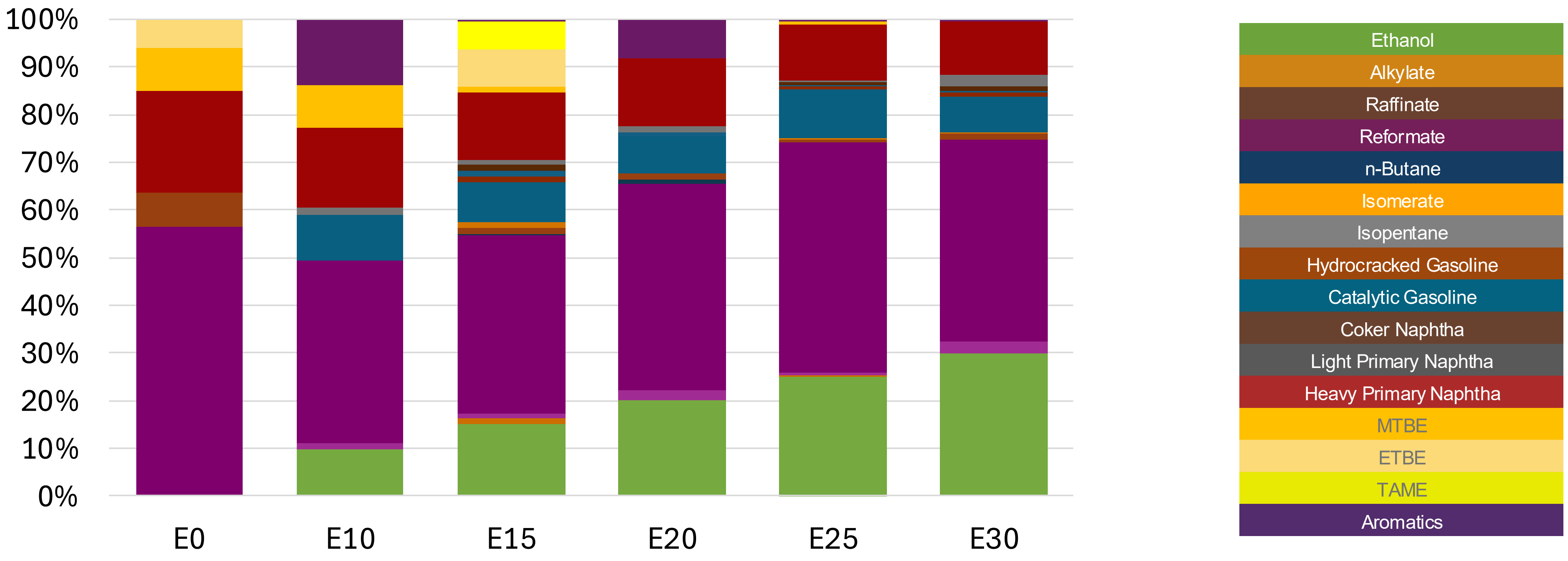


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.314	$ 2.197	$ 2.128	$ 2.056	$ 1.991	$ 1.941

Average CIF 2024 Prices

Source: Faro90

Bulgaria – Gasoline 95 – Increased Octane

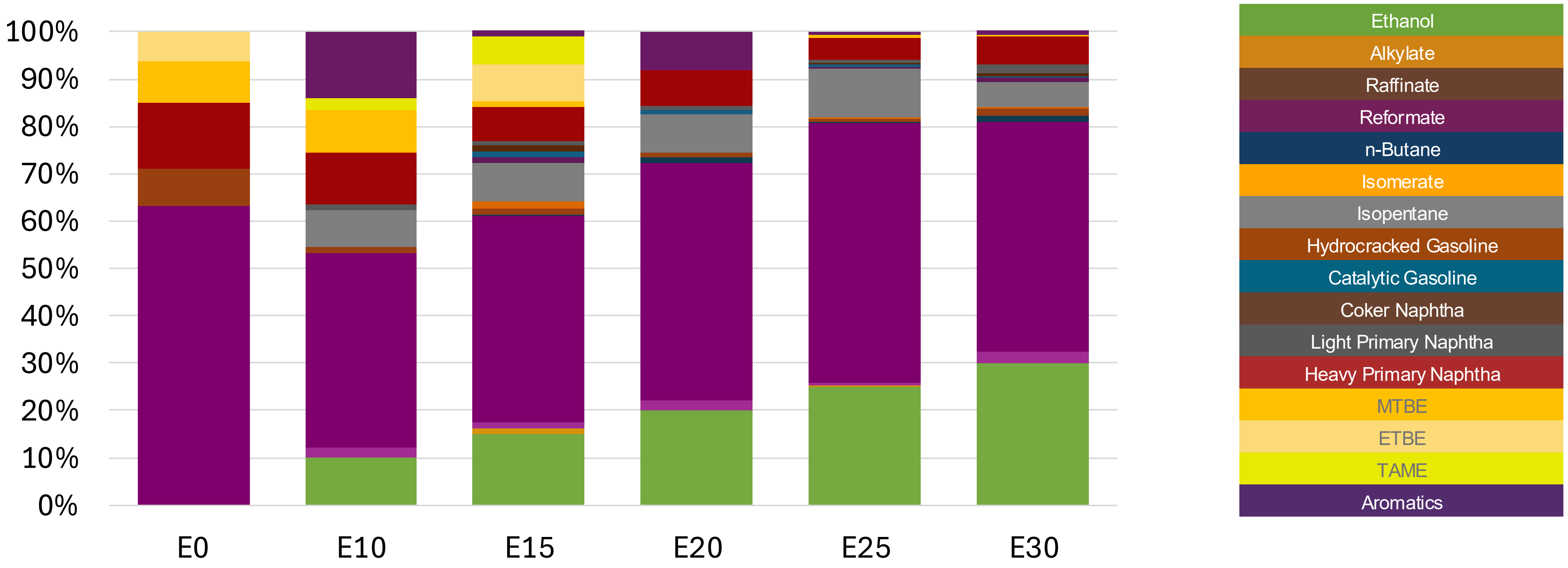


Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US$/gal)	$ 2.182	$ 2.182	$ 2.182	$ 2.182	$ 2.182	$ 2.182

Average CIF 2024 Prices

Source: Faro90

Bulgaria – Gasoline 98 – Increased Octane



Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US$/gal)	$ 2.314	$ 2.314	$ 2.314	$ 2.314	$ 2.314	$ 2.314

Average CIF 2024 Prices

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the **International Vehicle Emissions Model (IVE).**
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

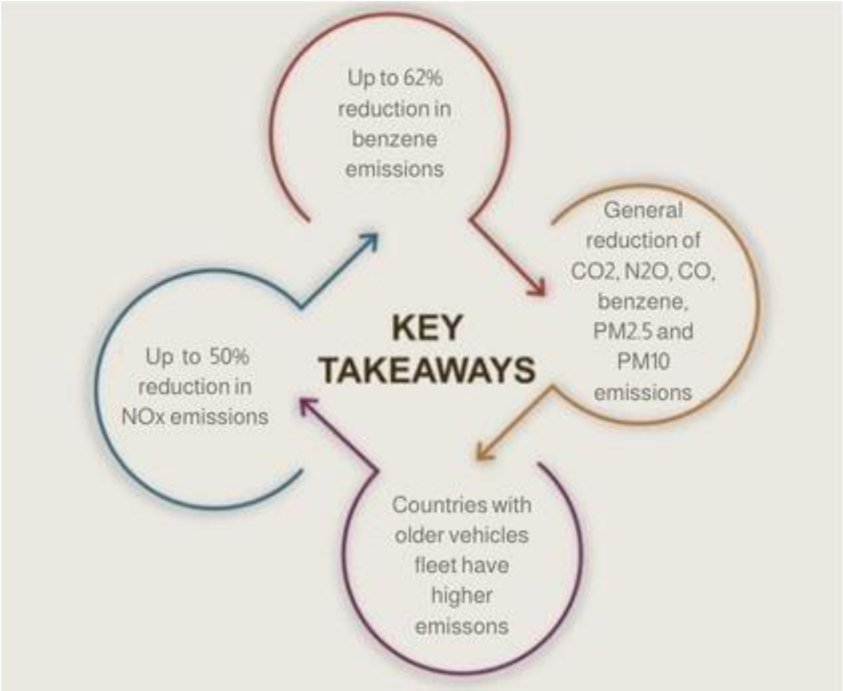
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Source: Bureau of transportation statistics.

Main Results



Gasoline Vehicle Fleet - Bulgaria

Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	1,016
		Single FI	3-Ways		>8 years	22,579
		Multi FI	3-Ways		>8 years	316,355
			Euro III		>8 years	485,118
			Euro IV		>8 years	419,978
			Euro V		>8 years	281,049
			Euro 6		>8 years	551,943
					4-8 years	121,575
					<4 years	316,881
			Hybrid		All	86,307
		Electric			All	22,574
	Medium	Carburator	3-Ways	PCV Tank	>8 years	100
		Single FI	3-Ways		>8 years	652
		Multi FI	3-Ways		>8 years	37,915
			Euro III		>8 years	92,509
			Euro IV		>8 years	151,461
			Euro V		>8 years	86,495
			Euro 6		>8 years	183,981
					[illegible]	[illegible]
					[illegible]	263,705
			Hybrid		All	67,747
		Electric			All	18,329
Sml Engines	Light	4-Cycle	None	PCV	>8 years	25,337
			Euro III		>8 years	36,683
			Euro IV		>8 years	156,292
			Euro V		>8 years	100,979
			Euro 6		>8 years	155,824
					4-8 years	178,140
					<4 years	391,093
			Hybrid		All	47,289
		Electric			All	12,556
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	16,574
		Fuel-Injection	3-Ways		>8 years	187,313
			Euro III		>8 years	105,052
			Euro IV		>8 years	121,096
			Euro V		>8 years	33,899
			Euro 6		>8 years	46,800
					4-8 years	22,064
					<4 years	2,867
			Hybrid		All	732
		Electric			All	2,005

3,368,231

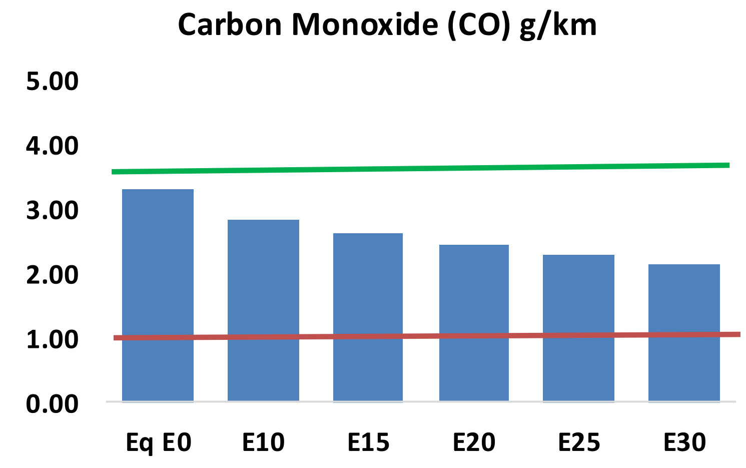
0 100 000 200 000 300 000 400 000 500 000 600 000

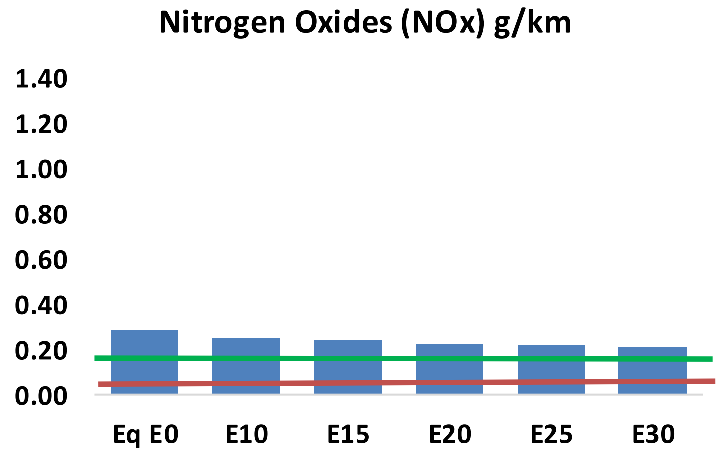
Vehicle Fleet: 5,239,744
Gasoline Vehicle Fleet: 3,368,231

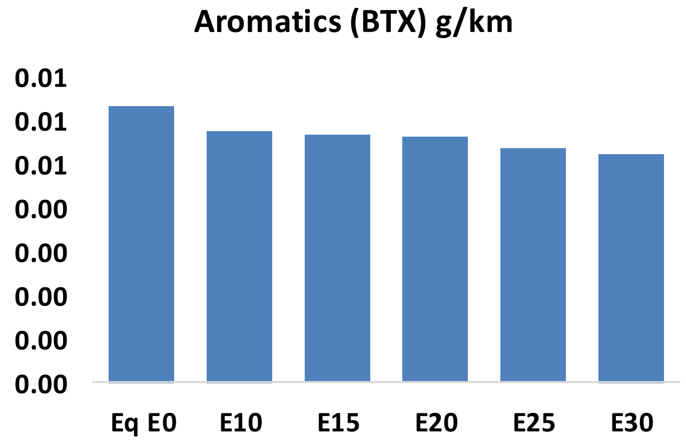
Source: Eurostat, ACEA

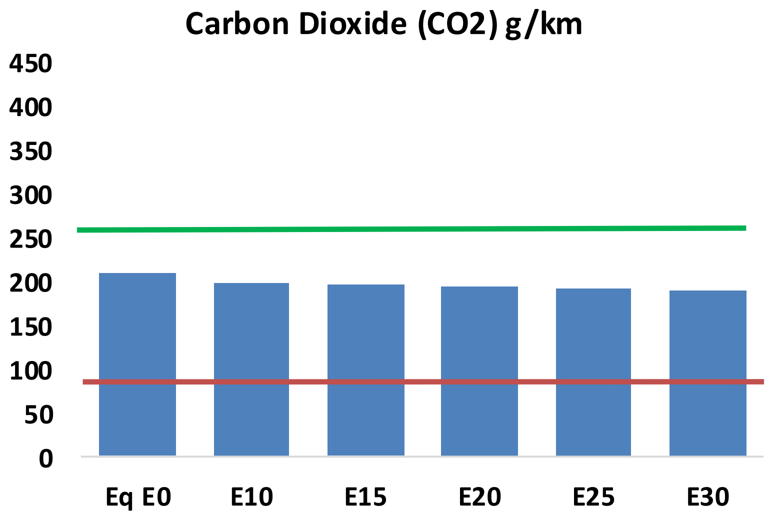
Bulgaria – Vehicular Emissions

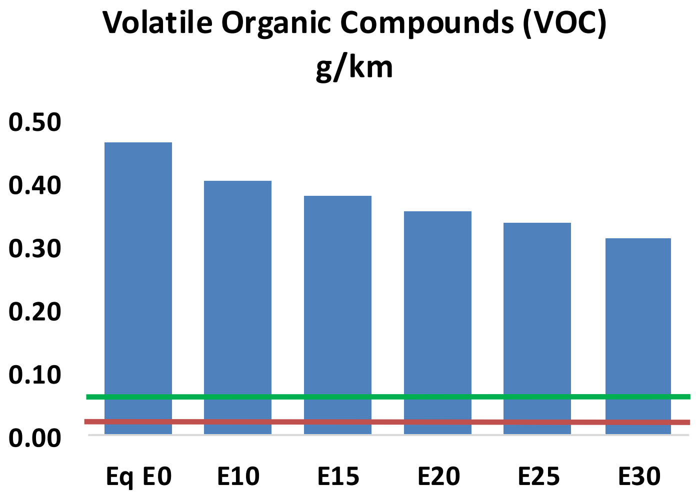
TIER III BIN 125 United States
Euro 6 Standard

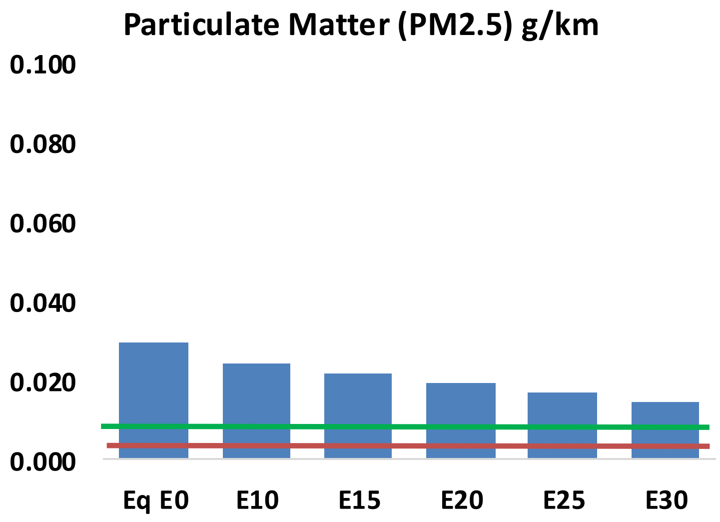












Source: Faro90

Bulgaria – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	263,442	244,624	225,807	210,667	195,560	183,983	171,087	-14%	-26%
CO2	16,709,105	16,279,442	15,873,649	15,554,505	15,396,581	15,309,889	15,027,684	-5%	-8%
VOC	37,213	34,755	32,297	30,361	28,450	26,991	25,005	-13%	-24%
NOx	22,626	20,891	19,701	19,094	18,117	17,352	16,450	-13%	-20%
SOx	190	176	162	149	138	125	112	-15%	-28%
NH3	5,912	5,854	5,795	5,823	5,888	5,934	5,972	-2%	0%
N2O	302	301	299	301	307	311	314	-1%	2%
CH4	8,123	8,123	8,123	8,245	8,423	8,578	8,724	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	270	261	252	247	243	240	235	-7%	-10%
Acetaldehyde	700	804	1,180	1,796	2,447	2,797	3,305	68%	249%
Formaldehyde	2,444	2,376	2,770	3,253	3,408	3,770	4,104	13%	39%
Benzene	507	487	461	454	450	431	417	-9%	-11%
PM2.5	979	890	801	723	646	564	478	-18%	-34%
PM10	1,363	1,239	1,115	1,006	899	785	665	-18%	-34%

Source: Faro90

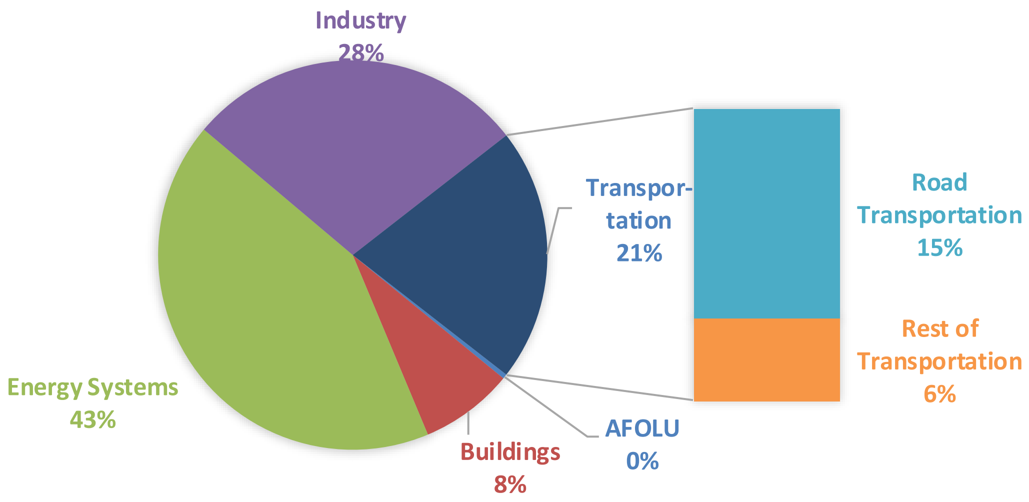
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

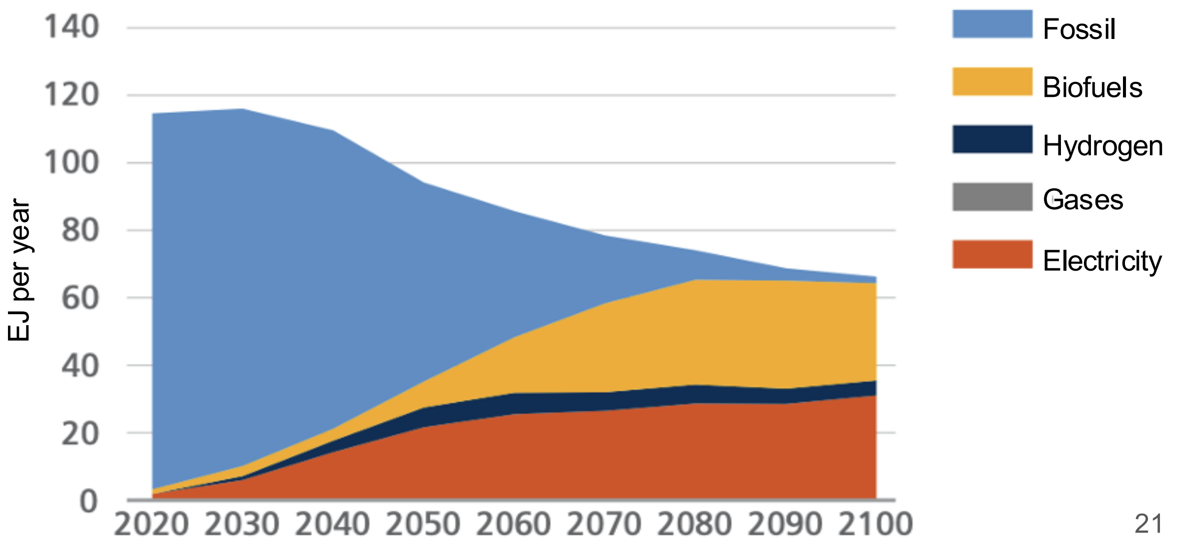
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE).** Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

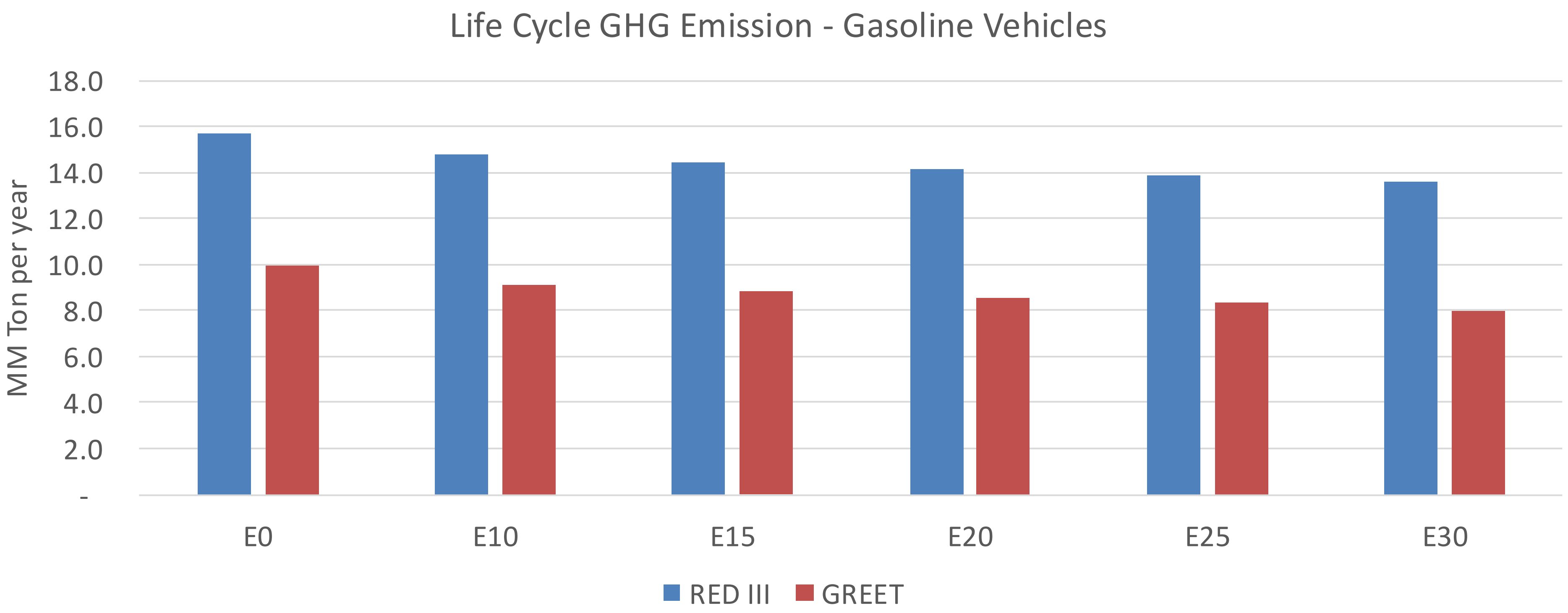
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Bulgaria – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	15.3
Target @ 2035 (MMT/year)	3.1
Delta	12.1

vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	8.2%	11.1%	13.8%	16.0%	19.2%
RED II/III	0.0%	5.7%	7.7%	9.6%	11.2%	13.4%

Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	6.7%	9.1%	11.3%	13.1%	15.7%
RED II/III	0.0%	7.4%	10.0%	12.5%	14.5%	17.3%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90